Intra-respondent reliability and enumerator quality in field conjoint experiments*

Matthew K. Ribar†

September, 2025

Abstract

Conjoint experiments ask respondents to consider multiple 'treatments' simultaneously, leading respondents to make errors in their responses. Field conjoint experiments introduce another error point because enumerators may make mistakes in administering the experiment. Calculating intra-respondent reliability (IRR) at the enumerator level provides researchers with a tool to monitor enumerator performance. Using related conjoint experiments in Côte d'Ivoire and Sénégal, I show that respondent characteristics fail to predict differences in switching rates between enumerators. Other common proxies for enumerator quality do not correlate with IRR. To account for variation in research contexts, researchers should normalize each enumerator's IRR using Z-scores. Including these calculations in high frequency data checks can proactively diagnose potential data quality issues.

Word Count: 1,918 (including references)

Key words: Conjoint experiments, survey methods, survey design

---

*This material is based upon work supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. DGE-1656518. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author and do not necessarily reflect the views of the National Science Foundation. The Senegalese data collection for this project was an output of the 'Structural Transformation and Economic Growth' (STEG), a programme funded by the Foreign, Commonwealth & Development Office (FCDO), contract reference STEG_LOA_1266_Ribar. Fieldwork in Cote d'Ivoire was supported by the Stanford King Center on Global Development, the Stanford Institute for Economic Policy Research, the Institute for Humane Studies, and a SurveyCTO data collection research grant. The Stanford University IRB approved the human subjects portion of this research under protocols number IRB-68012 and IRB-72215.

†Postdoctoral Fellow, Department of Political Science and the Weidenbaum Center on the Economy, Government, and Public Policy, Washington University in St. Louis mribar@wustl.edu

Researchers increasingly use field conjoint experiments to reach populations in developing countries. Garbe et al. (2024) use such an experiment to study willingness to register for a biometric ID program in Kenya. Baldin, Kao, and Lust (2025) use a field conjoint across Kenya, Malawi, and Zambia to identify when constituents are willing to entertain requests by formal or informal leaders. Zhou (2024) explores public preferences over territorial disputes with a conjoint experiment in India which asks opinions about Kashmir. Auerbach and Thachil (2018) study clientelism and brokers in Indian slums. In such surveys, research teams go to respondents and administer the survey in person, often using tablet computers.[1] In contrast to online conjoint experiments which use software like Qualtrics to engage directly with respondents, these tablet-based interviews require an enumerator to administer them. Adding an enumerator into the survey flow opens a space for enumeration errors to degrade survey quality (Adida et al. 2016; Di Maio and Fiala 2020).

I build on Clayton et al. (2023), who propose a statistical correction to account for switching error in conjoint experiments and introduce a measurement they call intra-respondent reliability to calculate it. Switching error occurs when respondents provide an answer to the conjoint experiment that does not reflect their true preferences. Conjoint experiments generally have multiple rounds; by adding an additional round which repeats or inverts a previous round, researchers can calculate the likelihood that respondents provide the same answer to identical conjoint rounds. Averaging this statistic at the enumerator level captures the fidelity with which the enumerators implement the conjoint.

Researchers who rely on enumerators to implement conjoint experiments should include enumerator-specific averages of intra-respondent reliability (IRR) in their regular quality monitoring to proactively diagnose problematic enumerators. Specifically, I recommend that researchers calculate normalize enumerator average IRR—calculated as the fraction of respondents who agree with themselves between rounds—using Z-scores. Normalizing the data helps to account for differences between specific surveys and research contexts. Researchers can flag enumerators whose average IRR exceeds a certain

[1] Ferree et al. (2023) embed a conjoint experiment into a phone survey in Malawi, showing that enumerator considerations in conjoint experiments are not exclusive to tablet-based field surveys.

threshold for additional supervision.

I illustrate the use of IRR as a tool to monitor enumerators through two related conjoint experiments I ran in Sénégal in 2023 and Côte d'Ivoire 2024. I presented respondents with two parties to a hypothetical land dispute and asked the respondent which profile was more likely to win. Profiles varied across the party's sex, the value of the party's land, the party's ethnicity (farmer or herder in Sénégal, autochthone or allochthone in Côte d'Ivoire), whether the party had given the traditional chief a gift, and whether the party possessed a written title for their land.[2] Using these data, I first use respondent's demographic information to predict their likelihood of switching (i.e. the probability that their response to the same question differs across rounds). I then show that enumerators are balanced across these predicted propensities to switch. Respondent characteristics do not predict IRR, but enumerator indicators do.

1 Intra-Respondent Reliability

The canonical conjoint experiment presents respondents with two profiles, side-by-side. Each profile has a set of attributes which randomly vary between different levels. Respondents are then asked to choose one of the two profiles. Common questions include "for which of these two candidates would you vote" or "which of these two policies do you prefer?" Each pair of profiles comprises one conjoint round. Researchers often show multiple rounds to increase statistical power. Bansak et al. (2018) show that increasing the number of conjoint rounds does not degrade conjoint quality or induce survey satisficing.

The conjoint design allows respondents to evaluate profiles across multiple attributes, but it also violates several common principles of survey design. Conjoint designs ask respondents to hold large amounts of information in their head and profiles combine multiple treatments arms.[3] These quirks of survey design make respondents vulnerable to 'switching error:' respondents may unwittingly choose profiles that do not correspond

[2] See Ribar (2025) for a full description of the experiments.

[3] Conjoint experiments are a sub-genre of factorial experiments (Hainmueller, Hopkins, and Yamamoto 2014.)

to their true preferences (Clayton et al. 2023).[4] This error can attenuate estimates and complicate sub-group analysis. To overcome this problem, Clayton et al. (2023) introduce a statistical correction for this source of bias for the two most common estimands for conjoint experiments: average marginal component effects (AMCEs) and marginal means.[5] These corrections rely on a quantity they call intra-respondent reliability (IRR): the likelihood a respondent makes an identical selection when faced with identical tasks.

The preferred manner in which to capture IRR is to repeat the same conjoint task at the beginning of the set of conjoint experiments and at the end. In my running example, the survey presented respondents with six total paired conjoint tasks, the sixth being an inverted copy of the first. We can then calculate IRR as the fraction of respondents who agreed with themselves across the two tasks. An IRR of 0.5 indicates that respondents agree with themselves 50 percent of the time—as randomly as if they flipped a coin. An IRR of one would indicate that respondents always provide identical answers to identical conjoint pairs. Online conjoint experiments average an IRR of about 0.77 (Clayton et al. 2023: 13); the in-person conjoint experiments in Sénégal and Côte d'Ivoire I explore below averaged an IRR of about 0.84.

2 Using Intra-Respondent Reliability to Monitor Enumerators

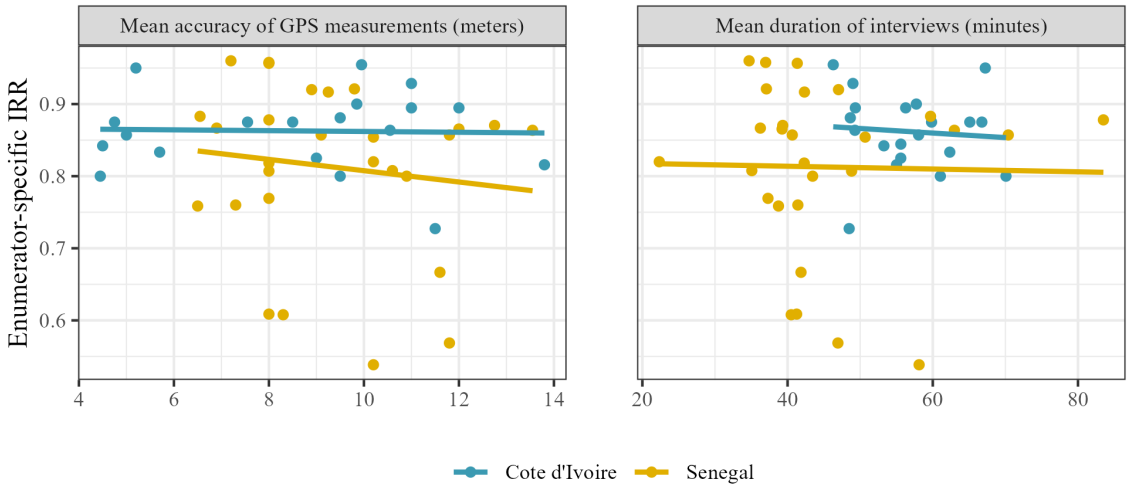
IRR is an excellent tool to monitor enumerator quality, beyond its role in correcting for measurement error. Monitoring the average IRR per enumerator allows survey researchers conducting high frequency data checks during data collection to identify which enumerators are effectively delivering the conjoint experiment's content and which enumerators are not.

Typically, researchers monitor enumerators and data quality through high-frequency data checks which produce a variety of diagnostic statistics to assess the data quality and identify problems. Such data checks can identify enumerators who shirk or who other-

[4] Another concern is that respondents will anchor themselves to the first or last attribute, which is why researchers commonly randomize the order in which attributes are presented.

[5] See Clayton et al. (2023: 11–2) for the formulae, which I omit here for brevity.

Figure 1. Proxies for enumerator shirking and enumerator-specific IRR are poorly correlated



The vertical axis of this figure shows enumerator-specific IRR; the horizontal axes show the mean accuracy of GPS coordinates (high values mean imprecise measurements) and the average survey duration, both calculated at the enumerator level. The lefthand panel excludes an extreme outlier in GPS accuracy that does not affect the overall conclusion. The correlation coefficient between enumerator IRR and duration is approximately 0.1; between enumerator IRR and accuracy it is approximately -0.01. Neither correlation is statistically significant.

wise do not collect high-quality data. To that end, many survey researchers monitor the duration of surveys (or the duration of specific modules within surveys) as part of these high-speed checks. Another common check is to use GPS coordinates to ensure enumerators are administering the surveys in the correct locations or following a sampling plan. While it is difficult to automate GPS based checks, many researchers also require a minimum accuracy threshold for any GPS measurements. More accurate GPS coordinates show that enumerators spent more time identifying their locations.

These tools detect enumerator shirking in general, but adding enumerator IRR to regular quality checks directly monitors the fidelity and effectiveness with which enumerators implement the conjoint experiment. Because of the complexities of the conjoint design, enumerators may successfully administer other parts of the survey but compromise data quality for the conjoint. Figure 1 demonstrates that enumerators who do not

rush through the survey and who take accurate GPS coordinates, two common metrics by which researchers monitor enumerators, may nevertheless do a poor job administering the conjoint.[6] Both mean duration and mean GPS accuracy are poorly correlated with IRR.

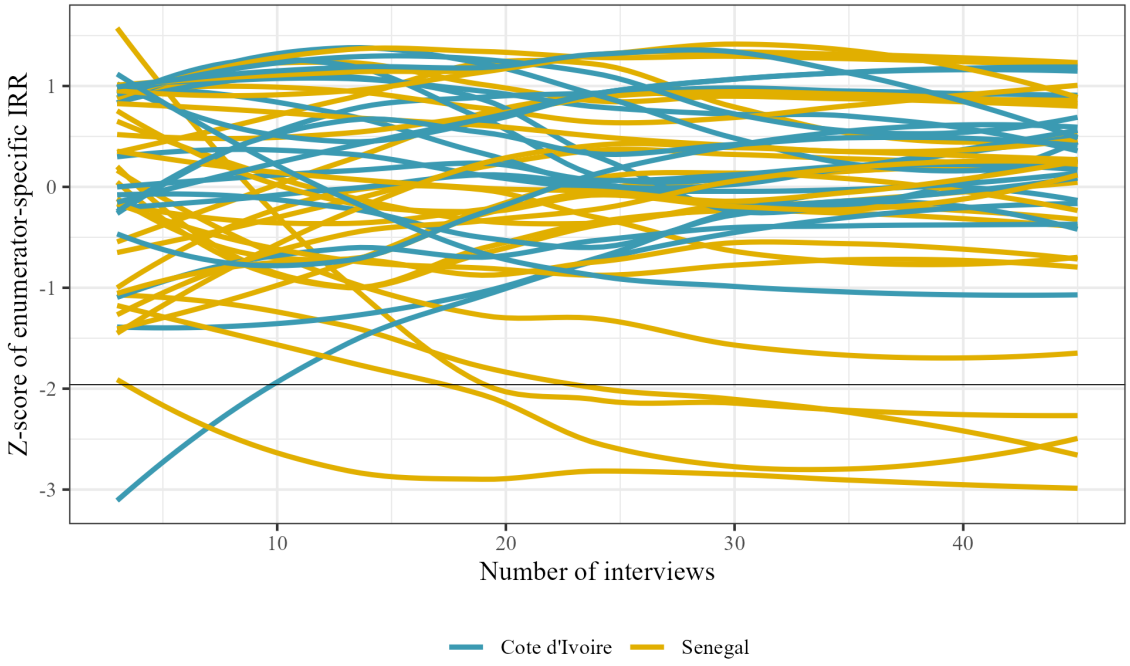
Figure 2 shows how enumerator specific IRR change over the period of data collection. This figure shows Z-scores of enumerator-specific IRR, after an enumerators n-th interview. So where the number of interviews (the horizontal axis) is ten, the values shows the the Z-scores of enumerator-specific IRR for all enumerator's first 10 interviews. In other words, it shows how far a given enumerator's average IRR is from other enumerator's average IRR. The figure shows that IRRs generally stabilize over time and the high performers (and low performers) become clear. Any given enumerator could encounter a series of inattentive respondents and therefore have comparatively low IRR, but figure 1, suggests that. after normalizing, such a series of respondent would not distort an enumerator's IRR.

Anecdotes from the conjoint experiments in Sénégal and Côte d'Ivoire evidence the utility of adding enumerator-specific IRR to high frequency data checks. In Côte d'Ivoire, I assigned an extra supervisor to oversee the enumerator with the lowest IRR. The supervisor discovered that—contra the survey training—the enumerator was reading conjoint profiles aloud to the respondent rather than showing the tablet screen, which displayed images of the conjoint profiles. Respondents surveyed by this enumerator were holding much greater amounts of information in their head, and as a result were making inconsistent selections. Alternative strategies to monitor enumerators would not have detected this problem, but checking enumerator IRR allowed me to correct the enumerator's behavior.

Likewise, in Sénégal, examining enumerator-specific IRR revealed a handful of enumerators with low IRRs. Enumerators for both conjoint experiments presented respondents with a printed guide for the different attribute levels, to help respondents keep track of the different information. After assigning additional supervision to these enumerators,

[6] It would be useful to compare IRR at the enumerator level to the duration of the conjoint module, to test whether enumerators with low IRR spent more or less time specifically administering the conjoint experiment. Unfortunately, I did not collect these data and am unable to make the comparison.

Figure 2. Enumerator-specific IRR smooths for enumerators over time



The vertical axis of this figure shows Z-scores of enumerator-specific IRR, updated after each interview. The horizontal axis shows the enumerators' count of completed interviews. While initial estimates of enumerator-specific IRR are noisy, they smooth after approximately 15 interviews.

the supervisors reported that enumerators were not using this printed guide. The supervisors corrected this error, and IRR improved. In both cases, including enumerator-specific IRR in regular data checks allowed me to identify errors which would not have been detected using other methods.

This approach is also more difficult for enumerators to "game" than alternative approaches to quality control such as adding attention checks. Across 1,965 unique respondents and 48 unique enumerators in my running example, no enumerator or respondent noticed that the first and sixth conjoint tasks were inverses of each other. If enumerators are unaware of this check, then they cannot correct respondents' answers. If enumerators were aware the researcher was checking for IRR, they would still have to keep the knowledge of the initial conjoint round in their head through all six rounds, which would

be non-trivial.[7]

Beyond enumerator monitoring, IRR could also be used for post-hoc corrections. One option is to drop enumerators with IRR below some pre-specified (and presumably pre-registered) threshold. Certainly, data should be dropped if an enumerator's IRR approaches 0.5—which would imply that the enumerator's respondent were answering as if randomly. The other option would be to use the correction proposed by Clayton et al. (Clayton et al. 2023). Using the correction results in a two step procedure for which it is challenging to estimate variance, so it is difficult to definitively say when it would be better to drop observations and when it would be better to use the correction. In the case of the two conjoint experiments in Sénégal and Côte d'Ivoire, dropping poor performing enumerators does not affect the statistical significance of my results.[8] However, in these surveys, I targeted enumerators with poor IRRs during the survey for increased supervision and coaching. In surveys where IRR was not monitored, dropping poor-performing enumerators may have a greater effect, but using enumerator-level IRR for correcting survey performance seems preferable to using it for post hoc corrections.

I recommend flagging enumerators after normalizing enumerator average IRR using Z-scores. Researchers could pre-specify a critical value beyond which they would flag enumerators for additional oversight or training. For example, the researcher could set a threshold of two standard deviations above the norm ($Z > 1.96$), and flag enumerators whose normalized IRR is beyond this threshold. Where the researcher believes that respondents are systematically different between enumerators, the researcher could estimate respondent level propensity to switch and flag enumerators based on the difference between expected and actual switching. Respondent characteristics fail to predict switching in this context, but easily could elsewhere. Section 3 uses a similar approach to demonstrate that enumerators, not respondents' demographic questions, drive the variation in enumerator-average IRRs.

[7] If enumerators seem to be gaming the system, more complicated alternatives could involve randomizing whether the IRR check is actually inverted or alternating between the fifth and sixth conjoint rounds containing the repeated profiles to further complicate enumerator malfeasance.

[8] Specifically, I test the difference in marginal means between two subgroups: the lowest quartile of trust in formal institutions and the highest quartile of trust in formal institutions. Dropping 1-3 enumerators does not affect this t-statistic; dropping more reduces it due to the decrease in sample size.

3 What predicts IRR?

Are low enumerator-specific IRRs due to enumerators or respondents? Enumerators who are going an excellent job may nevertheless face inattentive respondents. Because enumerators are generally assigned geographically—one team implements the survey in region A, another in region B—respondent characteristics such as education or occupation could be systematically different across enumerators. If education predicts a respondent's IRR and average education of respondents differs across enumerators, then enumerator IRRs would not be an accurate way to monitor enumerator performances.

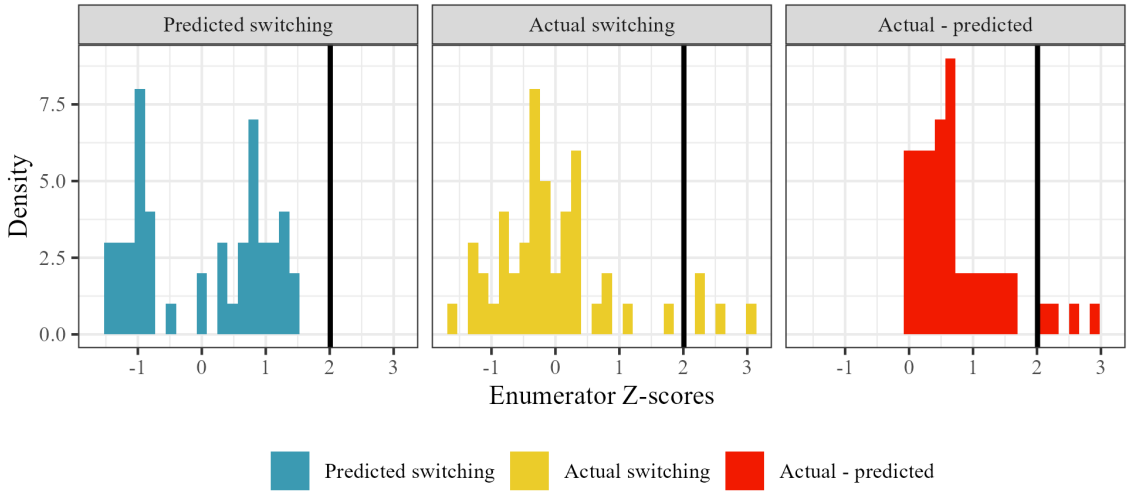
To identify the extent of this problem, I use a logit model to predict the likelihood of a respondent switching their responses.[9] The outcome is a binary indicator of switching (i.e. what I average to find enumerator-specific IRR). The predictors are the respondent's demographics: age group indicators, education, sex, and the respondent's relationship to the head of the household. I also include the country in which the survey took place. Figure 3 shows these results in three ways. The lefthand panel shows the distribution of the predicted propensities to switch; the center panel shows the distribution of switching itself; and the righthand panel shows the distribution of the absolute value of error (i.e the absolute value of the actual switching minus the estimated propensity to switch). For all statistics I first convert to enumerator averages and then to z-scores. A value to the right of the vertical line indicates a statistically significant outlier.

The lefthand panel of figure 3 shows that the predicted propensity of switching error is reasonably balanced among enumerators; there is no enumerator for whom the average predicted propensity to switch is more than two standard deviations away from the mean. On the other hand, the central panel shows four enumerators with high values of average switching error (the same four enumerators that figure 2 identifies). The righthand panel shows that these outliers persist even when accounting for respondent's propensity to switch.

More formally, a joint-F test of enumerator indicators rejects the null hypothesis that

[9] I also predicted switching error using a linear probability model (LPM) and a random forest, as well as weighted ensemble of the logit, the LPM, and the random forest. These alternatives produced no appreciable improvement on prediction errors (measured through root mean squared error) over the logit in out-of-sample predictions.

Figure 3. Average switching and average predicted propensity to switch by enumerators



All statistics are averaged at the enumerator level and transformed into z-scores. Predictions are from a logit regression of demographic variables on binary switching error at the respondent level. The vertical lines show critical values for a t-distribution with 48 degrees of freedom and $\alpha = 0.05$.

enumerator indicators fail to predict IRR ($F = 1903$, $p < 0.001$). Demographic factors, on the other hand, do not jointly predict IRR either by themselves or when added to a model controlling for enumerators. These results support the claim that enumerators, not respondents, are largely responsible for differences in IRR levels across enumerators.

4 Conclusion

This research note advocates for using enumerator-specific IRR as a tool to monitor enumerator quality in field conjoint experiments. IRR captures whether the respondent provides identical answers to identical questions posed at the start and end of the conjoint experiment. Averaging these values at the enumerator level and normalizing via Z-scores can help researchers to monitor whether enumerators are correctly implementing the conjoint experiments. Respondent characteristics fail to meaningfully predict IRR, suggesting that enumerators drive deviations in IRR, rather than respondents. This ex-

tension of Clayton et al. (2023)'s results will be relevant to any survey researchers who implement conjoint experiments in contexts which require enumerators to administer the survey.

References

Adida, C. L., Ferree, K. E., Posner, D. N., and Robinson, A. L. (2016), "Who's Asking? Interviewer Coethnicity Effects in African Survey Data", *Comparative Political Studies*, 49 (12): 1630–60.

Auerbach, A. M., and Thachil, T. (2018), "How Clients Select Brokers: Competition and Choice in India's Slums", *American Political Science Review*, 112 (4) (): 775–91.

Baldwin, K., Kao, K., and Lust, E. (2025), "Is authority fungible? Legitimacy, domain congruence, and the limits of power in Africa", *American Journal of Political Science*, 69 (1): 314–29.

Bansak, K., Hainmueller, J., Hopkins, D. J., and Yamamoto, T. (2018), "The Number of Choice Tasks and Survey Satisficing in Conjoint Experiments", *Political Analysis*, 26 (1) (): 112–9.

Clayton, K., Horiuchi, Y., Kaufman, A., King, G., and Komisarchik, M. (2023), "Correcting Measurement Error Bias in Conjoint Survey Experiments", Working Paper.

Di Maio, M., and Fiala, N. (2020), "Be Wary of Those Who Ask: A Randomized Experiment on the Size and Determinants of the Enumerator Effect", *The World Bank Economic Review*, 34 (3): 654–69.

Ferree, K. E., Honig, L., Lust, E., and Phillips, M. L. (2023), "Land and Legibility: When Do Citizens Expect Secure Property Rights in Weak States?", *American Political Science Review*, 117 (1): 42–58.

Garbe, L., McMurry, N., Scacco, A., and Zhang, K. (2024), "Who Wants to be Legible? Digitalization and Intergroup Inequality in Kenya", *Comparative Political Studies*, 58 (9): 1803–53.

Hainmueller, J., Hopkins, D. J., and Yamamoto, T. (2014), "Causal Inference in Conjoint Analysis: Understanding Multidimensional Choices via Stated Preference Experiments", *Political Analysis*, 22 (1): 1–30.

Ribar, M. (2025), "Land, Power, and Property Rights: The Political Economy of Land Titling in West Africa", Doctoral Dissertation (Palo Alto, CA: Stanford University).

Zhou, A. (2024), "The Cost of Compromise: Territorial Disputes and Domestic Public Opinion".